Generative Image Modelling on Cat Images

Project III — Generative Models

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1 Project Overview

In this project, we investigate how different lightweight generative modelling approaches perform on the task of generating cat images. We compare three models: a GAN trained from scratch, an adversarial autoencoder, and a VQ-VAE.

Our experimental pipeline is organized into four phases. First, we prepare a fixed subset of the required Cat Dataset and train a GAN baseline from scratch. Then, we train an adversarial autoencoder and a VQ-VAE on the same data. Finally, we compare the generated images quantitatively using Fréchet Inception Distance and qualitatively through visual inspection and latent-space interpolation.

Our main priority is to ensure a fair comparison between the selected generative approaches rather than achieving the highest possible image quality. Since training generative models can be computationally expensive, we will train and evaluate the models on fixed subsets of the required datasets.

2 Dataset and Models

We will use the **Cat Dataset** from Kaggle, which contains 9,993 cat images with different resolutions and visual conditions. Since the images have different sizes, we will preprocess them to a fixed resolution before training.

For the main experiments, we will use:

Table 1: Dataset splits for main experiments.

Split / purpose	**Size**
Training subset	3,000 cat images
FID real reference set	1,000 separate cat images
Generated samples for FID	1,000 images per model
Unused / reserve images	5,993 cat images

The training subset and FID reference subset will be fixed across experiments to keep the comparison consistent.

All three models are trained from scratch on the cat subset; Conv-AAE-64 may alternatively be fine-tuned if a suitable pretrained checkpoint is available.

For all three models, we will primarily use images resized to 64×64 pixels. If training becomes too slow, we will reduce the resolution to 32×32 pixels. This keeps the comparison computationally feasible and ensures that all models are trained under similar conditions.

Table 2: Model architectures used in our experiments.

Model	Type	Size	Purpose
DCGAN-64	GAN	$\sim$3–5M	Baseline comparison
Conv-AAE-64	Adversarial autoencoder	$\sim$2–6M	Latent-space generation
VQ-VAE-64	Vector-quantized VAE	$\sim$5–15M	Discrete-latent generation

3 Experiments

3.1 Phase 1: GAN Baseline from Scratch

We will train DCGAN-64 from scratch on the 3,000-image cat subset. The model will consist of a convolutional generator and a convolutional discriminator.

We will investigate a small set of training settings:

Table 3: GAN hyperparameter search space.

Hyperparameter	Planned values / options
Latent vector size	64, 128
Learning rate	0.0002, 0.0001
Batch size	32, 64
Image resolution	64×64, fallback 32×32

We will monitor generator loss, discriminator loss, generated samples, diversity, and signs of mode collapse. If mode collapse occurs, we will try practical stabilization techniques such as lowering the learning rate, applying label smoothing, adjusting the discriminator-generator balance, or selecting an earlier checkpoint.

3.2 Phase 2: Adversarial Autoencoder

We will train or fine-tune Conv-AAE-64 on the same 3,000-image cat subset.

The AAE will consist of:

- convolutional encoder,
- continuous latent space,
- convolutional decoder,
- latent discriminator.

The AAE allows us to compare the GAN with a latent-space generative model. It is also useful for interpolation, because generated images can be directly linked to latent vectors.

We will evaluate reconstruction quality, generated samples, latent-space smoothness, and FID.

3.3 Phase 3: VQ-VAE

We will train VQ-VAE-64 on the same 3,000-image cat subset.

Table 4: AAE hyperparameter search space.

Hyperparameter	Planned values / options
Latent dimension	64, 128
Learning rate	10^{-4}, 10^{-5}
Batch size	32, 64 if possible
Image resolution	64×64, fallback 32×32

VQ-VAE is an extension of the variational autoencoder that uses a discrete latent codebook instead of a continuous latent distribution. This gives us a third lightweight generative approach without introducing the computational cost of diffusion models.

The model will consist of:

- convolutional encoder,
- vector-quantized latent codebook,
- convolutional decoder.

Table 5: VQ-VAE hyperparameter search space.

Hyperparameter	Planned values / options
Codebook size	128, 512
Embedding dimension	64, 128
Learning rate	10^{-4}, 2×10^{-4}
Batch size	32, 64
Image resolution	64×64, fallback 32×32

For VQ-VAE generation, we will decode sampled latent codes based on the empirical distribution of codes observed in the training subset. If time allows, we may train a small prior over the discrete latent space, but this will not be required for the main comparison.

3.4 Phase 4: Evaluation and Latent Interpolation

We will evaluate all models quantitatively and qualitatively.

For quantitative evaluation, we will compute Fréchet Inception Distance between:

1. 1,000 real images from the fixed FID reference subset;
2. 1,000 generated images from each model.

Table 6: Planned quantitative comparison.

Model	Training data	Generated samples	Metric
DCGAN-64	3,000 cats	1,000	FID
Conv-AAE-64	3,000 cats	1,000	FID
VQ-VAE-64	3,000 cats	1,000	FID

For qualitative evaluation, we will compare generated image grids and discuss:

- cat-like appearance;

- sharpness;
- diversity;
- artifacts;
- mode collapse;
- blurry outputs;
- training stability.

We will also perform latent interpolation. We will select two generated images together with their latent vectors and linearly interpolate between them to generate 8 additional images. This will produce 10 images in total: 2 endpoint images and 8 interpolated images.

The interpolation experiment will primarily use the GAN or AAE, because these models have the clearest latent representation.

4 Cats-and-Dogs Extension

As an exploratory extension, we will train or adapt one selected model on the Dogs vs. Cats dataset from the project requirements.

We will use a balanced subset:

Table 7: Dataset splits for cats-and-dogs extension.

Split / purpose	**Size**
Mixed training subset	1,000 cats + 1,000 dogs
Optional reference set	500 cats + 500 dogs
Generated samples	1,000 images

The goal is to compare the cat-only model with a model trained on both cats and dogs. We will assess whether the generated observations resemble cats and dogs as separate concepts or become ambiguous mixtures.

Since class-distinct generation can be difficult without conditioning, this part will be treated as exploratory and will not receive disproportionate training time.

5 Division of Work

Table 8: General division of responsibilities.

Area	**Mateusz Iwaniuk**	**Adam Kaniasty**
Project setup	Dataset download, preprocessing, environment	Experiment structure, logging, reproducibility
Phase 1	GAN architecture and training setup	GAN hyperparameter variants and mode collapse analysis
Phase 2	AAE implementation and training	AAE generation and latent-space analysis
Phase 3	VQ-VAE implementation and training	VQ-VAE sampling and codebook analysis
Phase 4	Qualitative comparison and visualizations	FID computation and quantitative analysis
Extension	Cats-and-dogs dataset preparation	Mixed-dataset training and comparison
Analysis	Interpretation of generated samples	Metrics, plots, and final discussion